



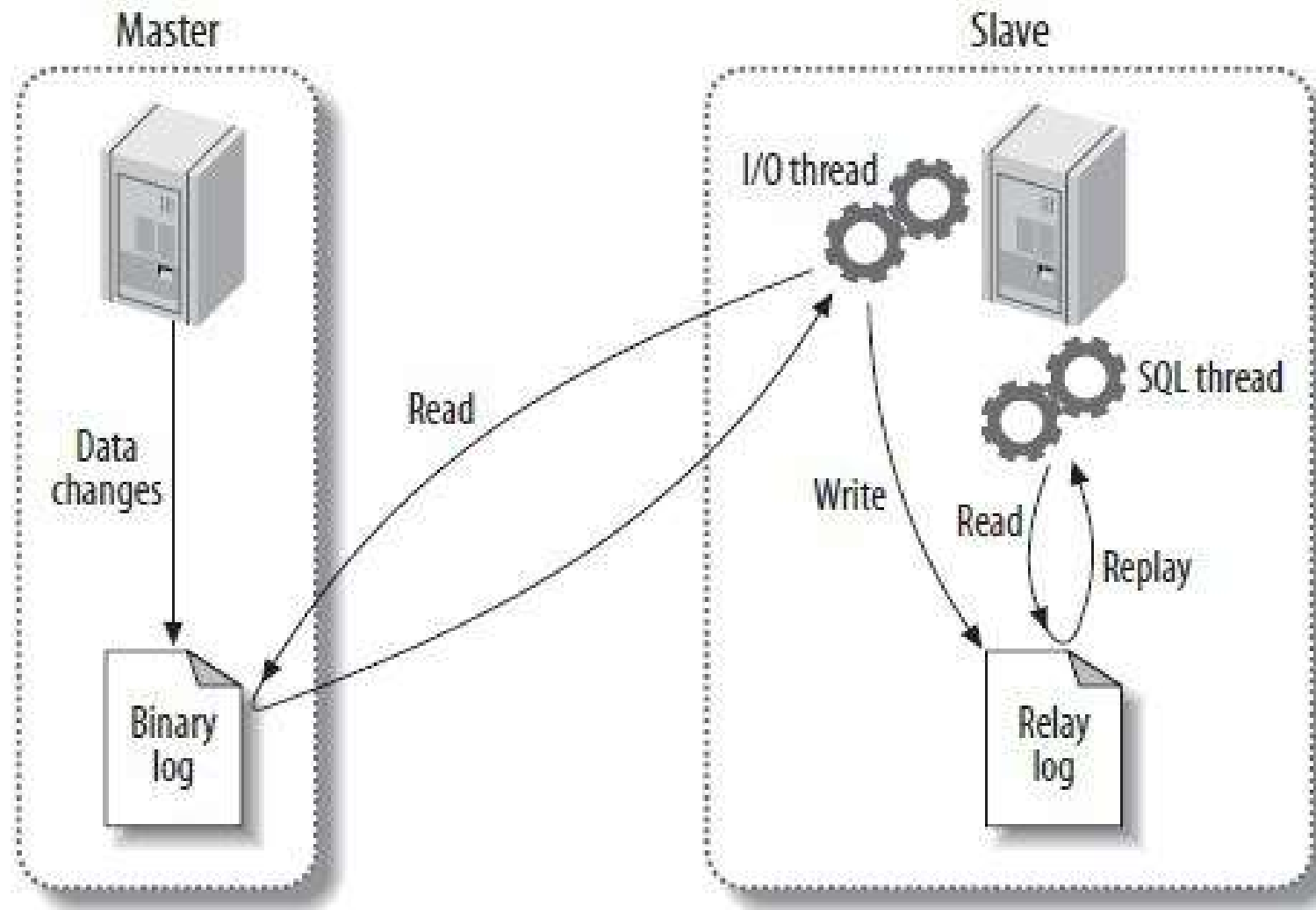
MySQL HA Solutions



- **MySQL Replication**
- **MySQL Replication w/Linux HA**
- **MySQL Replication w/Galera**
- **MySQL Cluster**



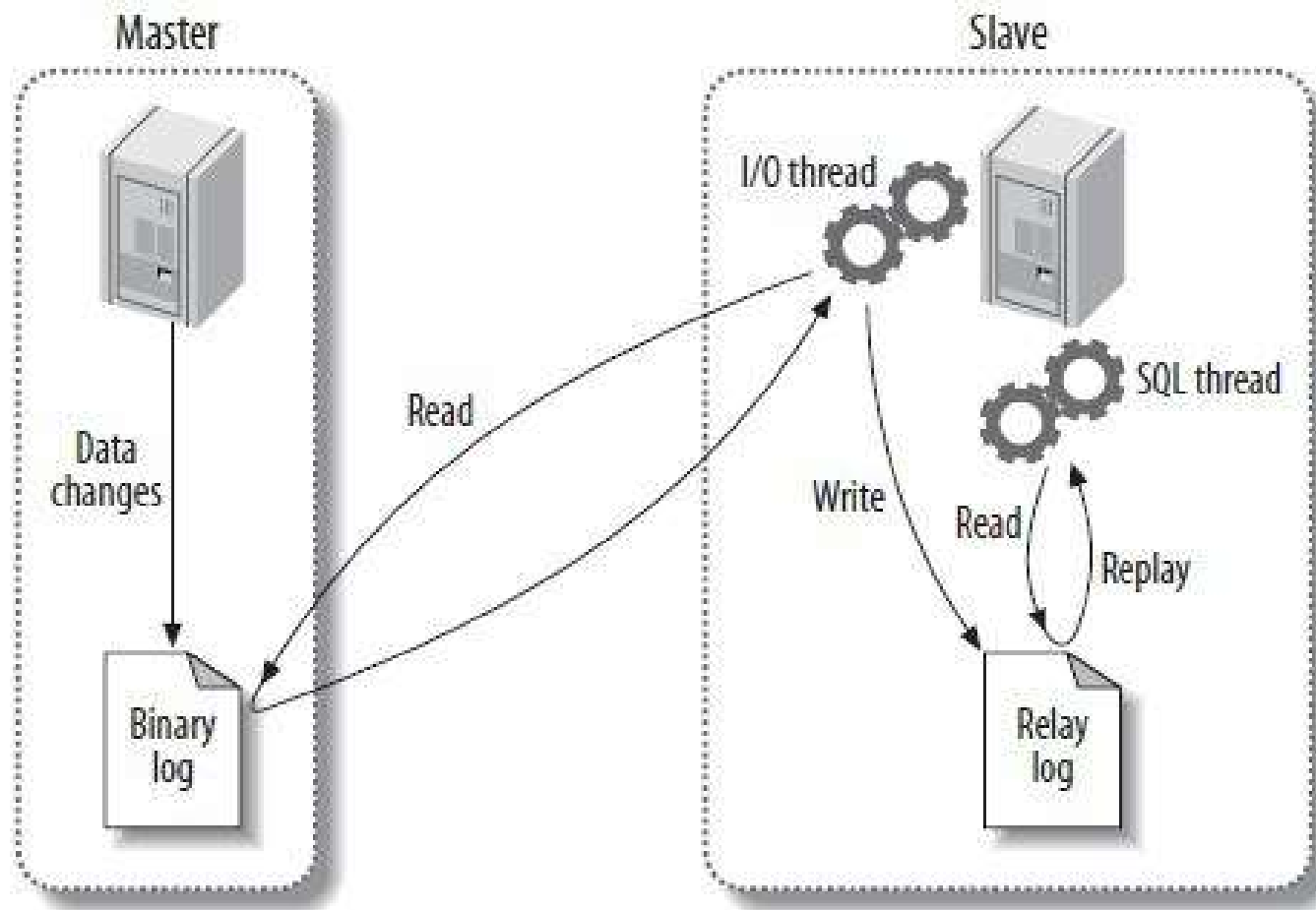
MySQL Replication M-S



Unidirectional

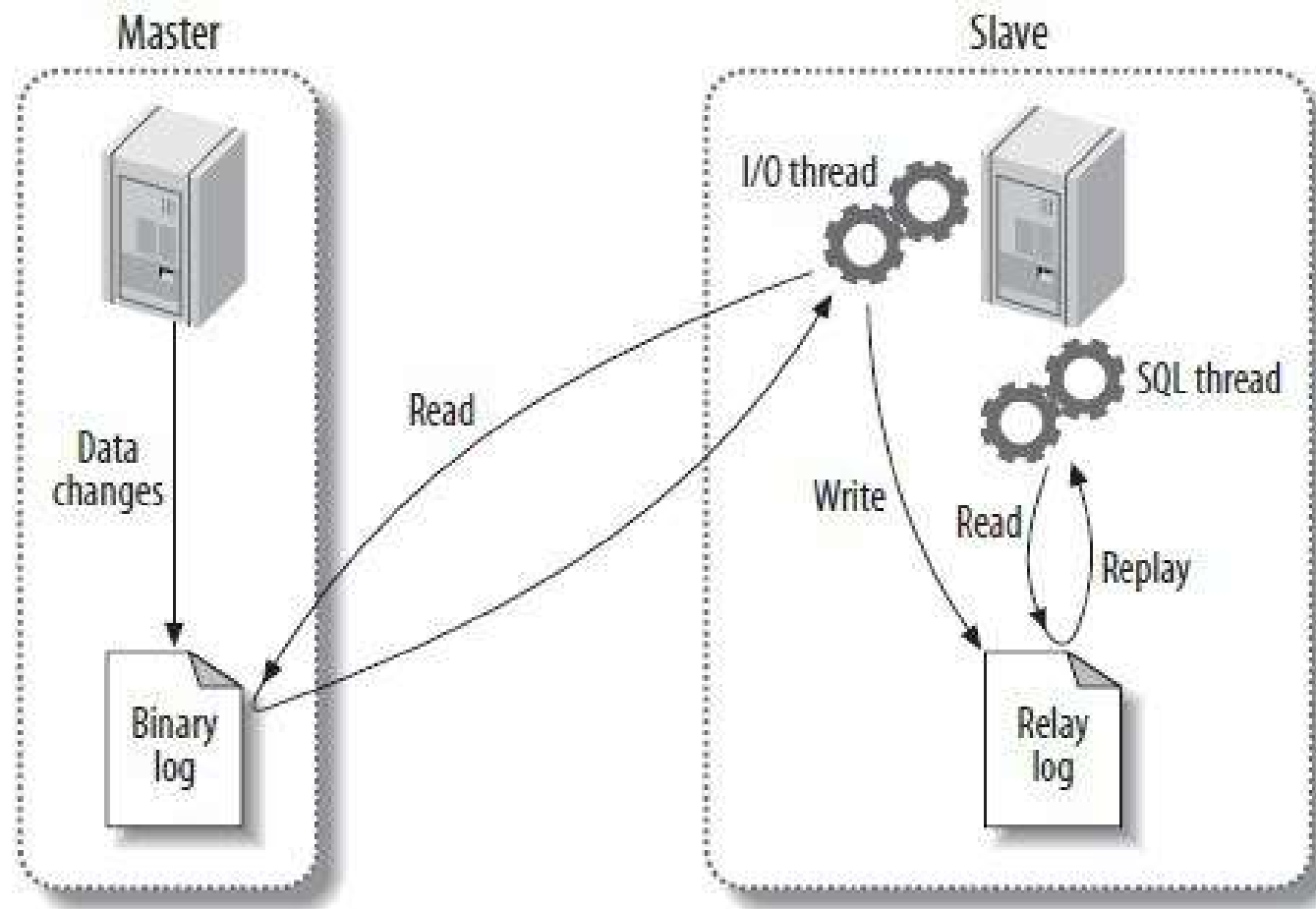


MySQL Replication M-M





MySQL Replication M-M

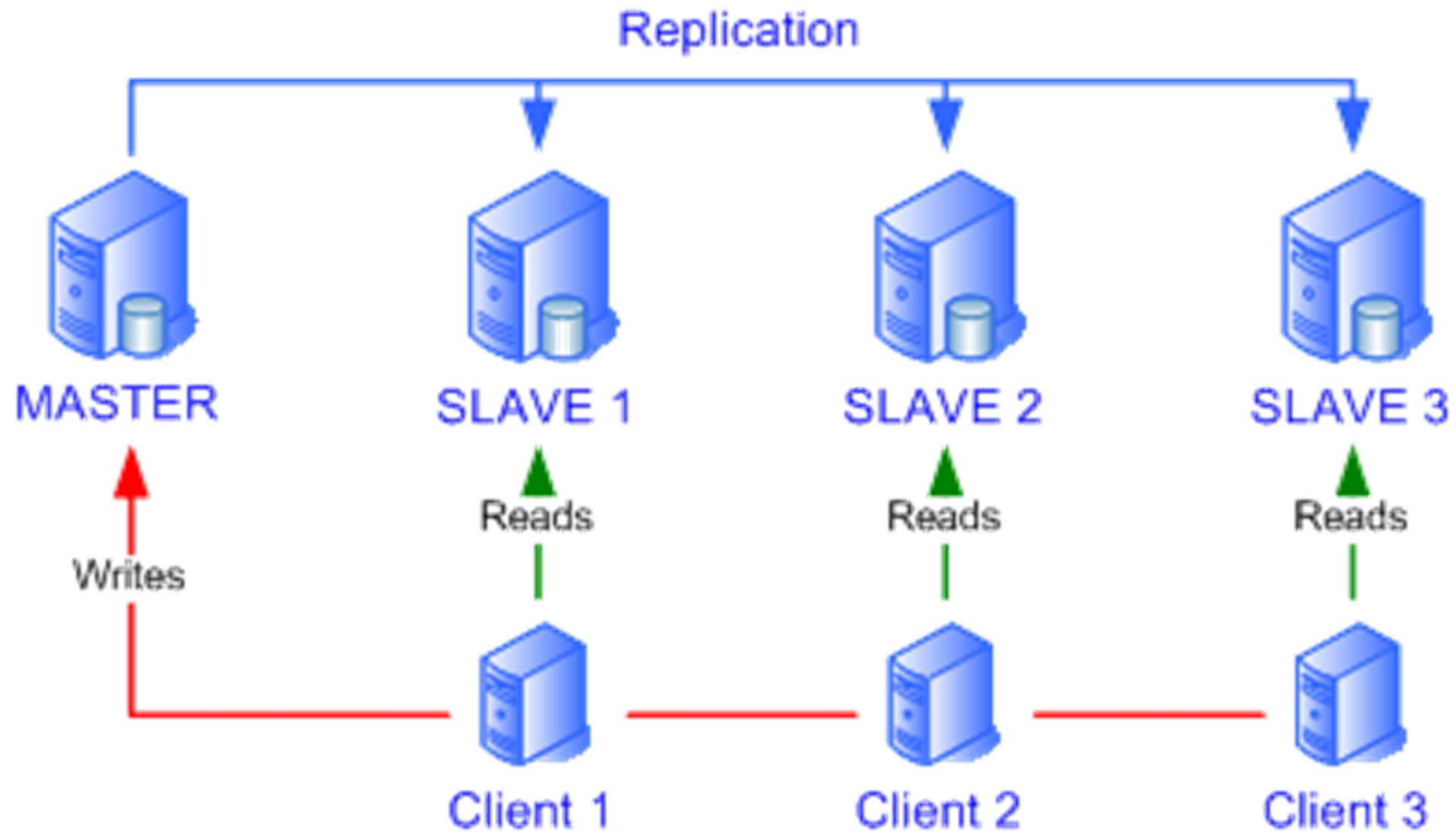


It is not a cluster is just a replication

(e.g. INSERT / DELETE)



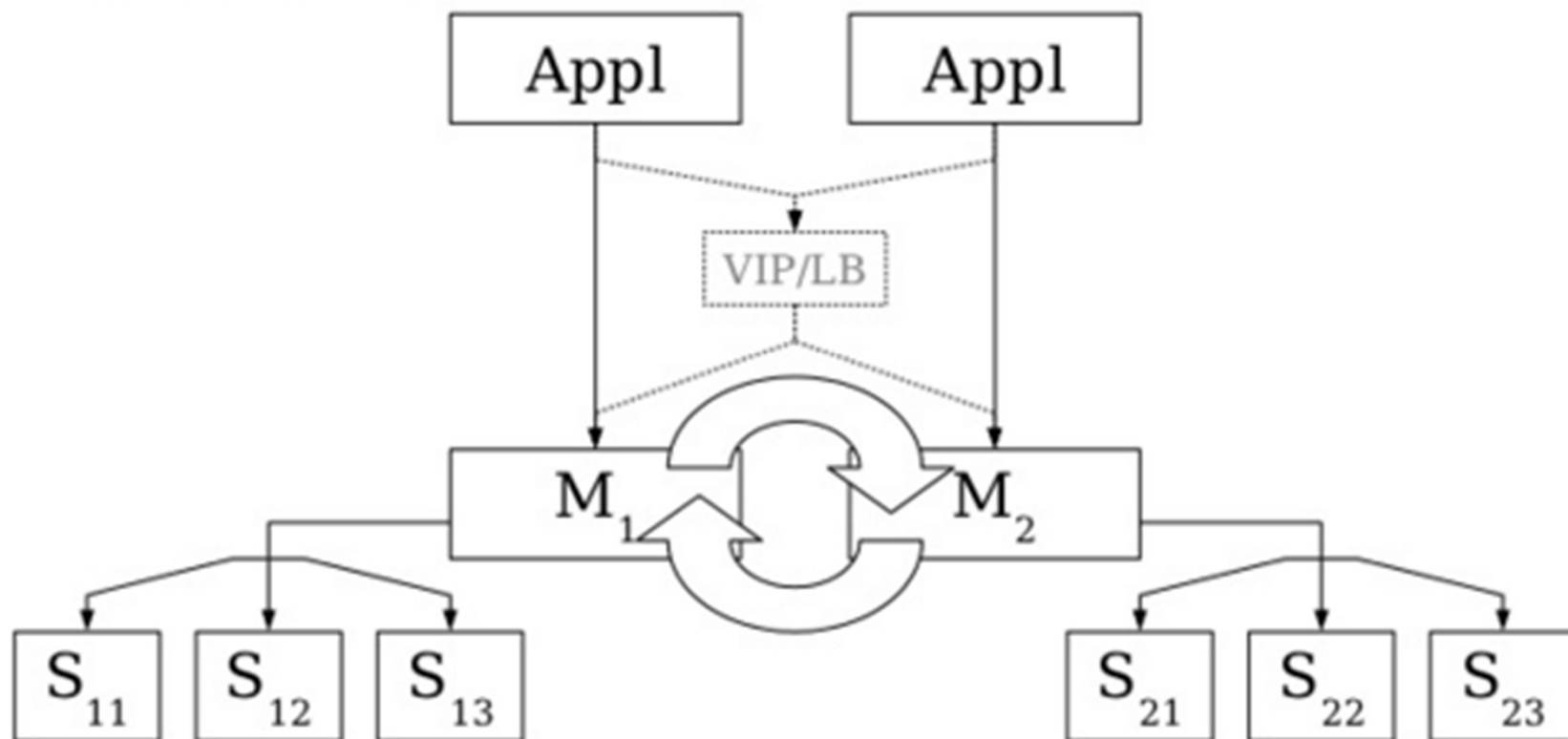
MySQL Replication M-S





MySQL Replication M-M

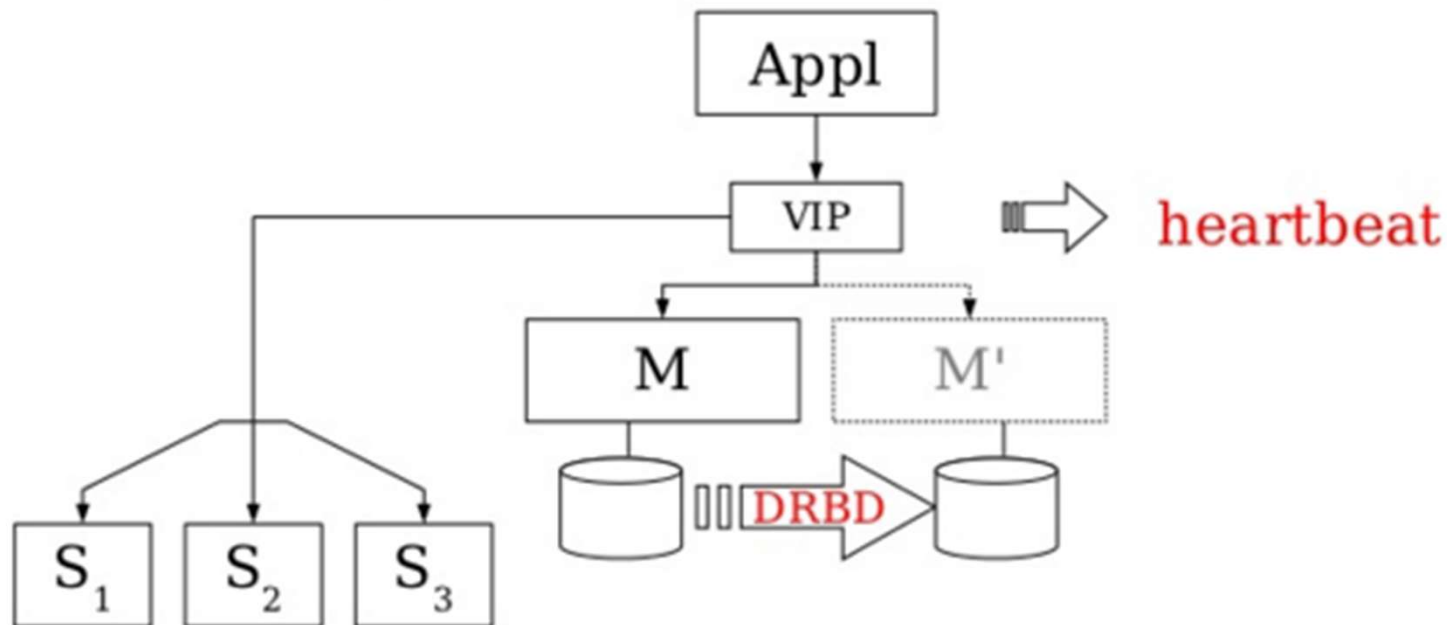
- actually Master-Master Replication





MySQL Replication & Linux HA

- HA with MySQL:

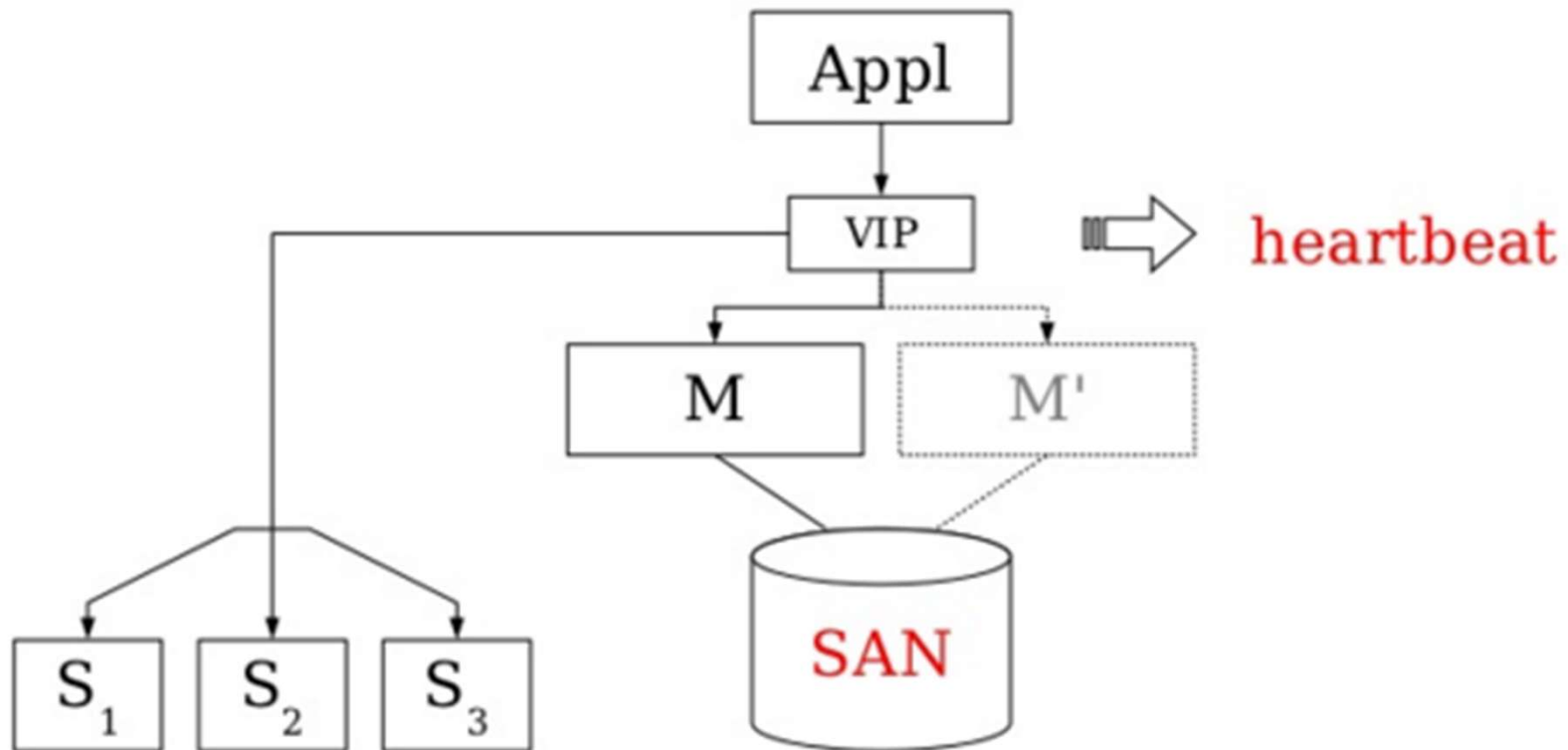


- 99.99% HA (four nine)
- active – passive fail over!



MySQL Replication & Linux HA

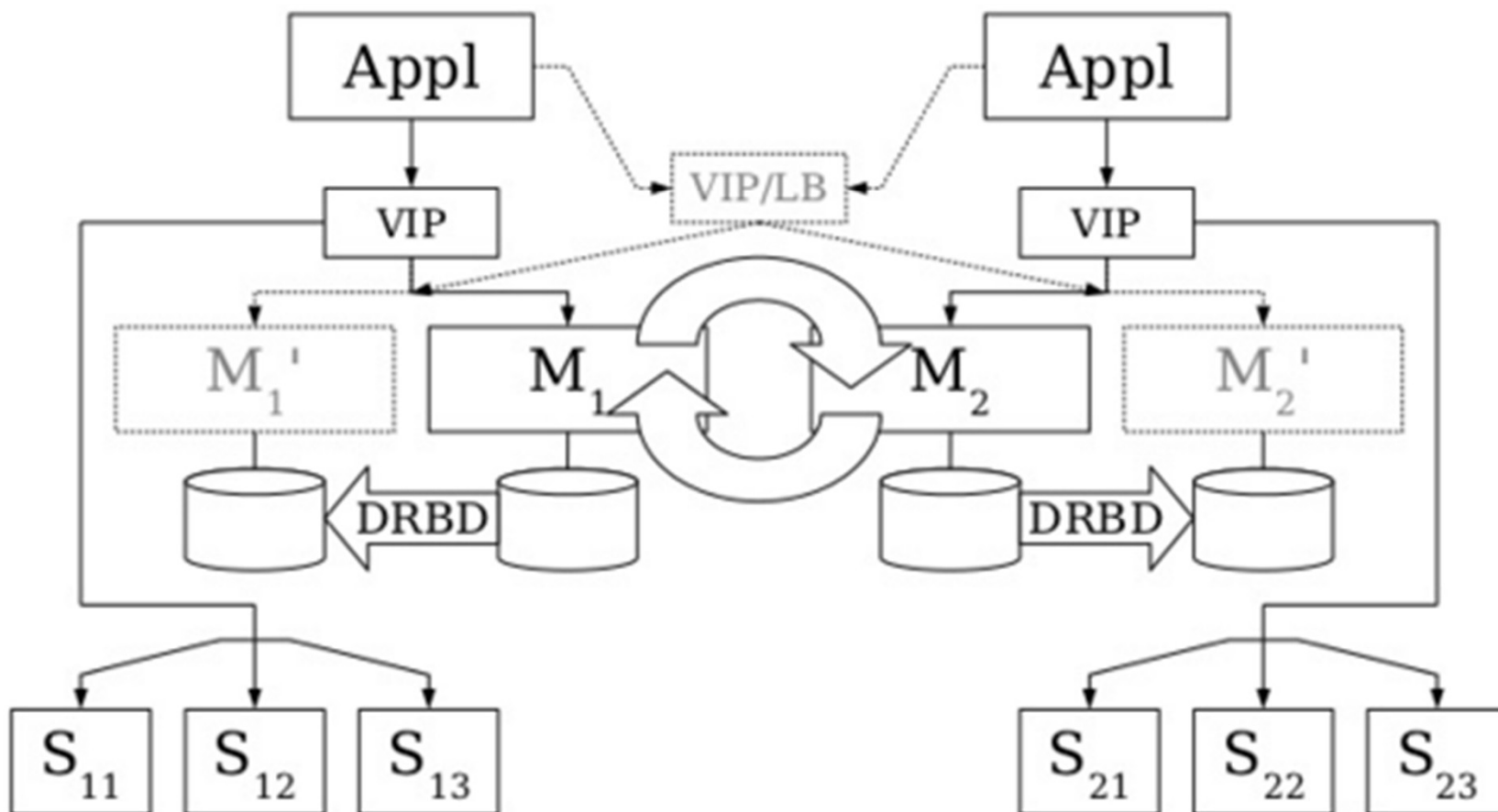
- HA with MySQL and SAN:





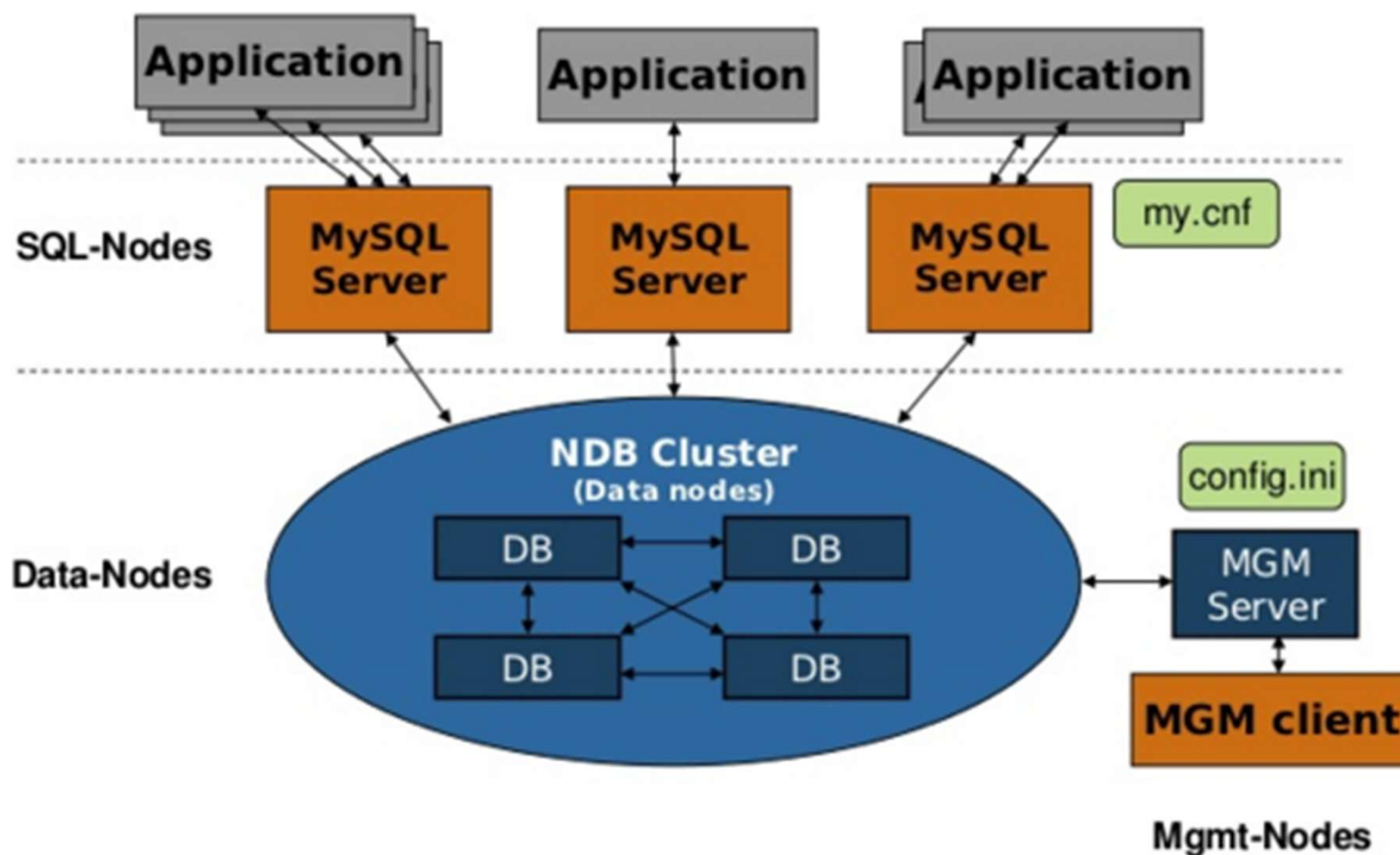
MySQL Replication & Linux HA

- HA Multi-Master Replication



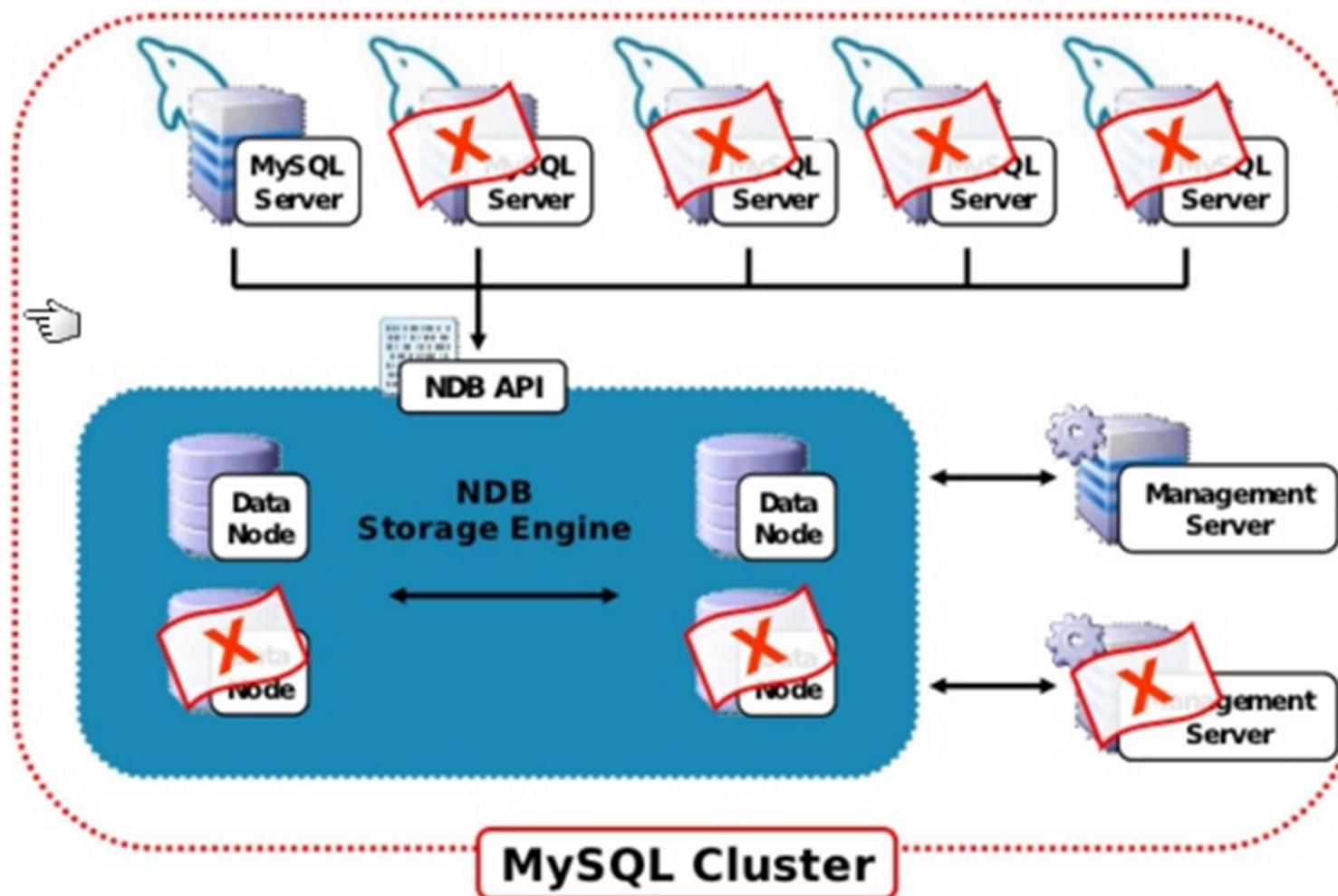


MySQL Cluster



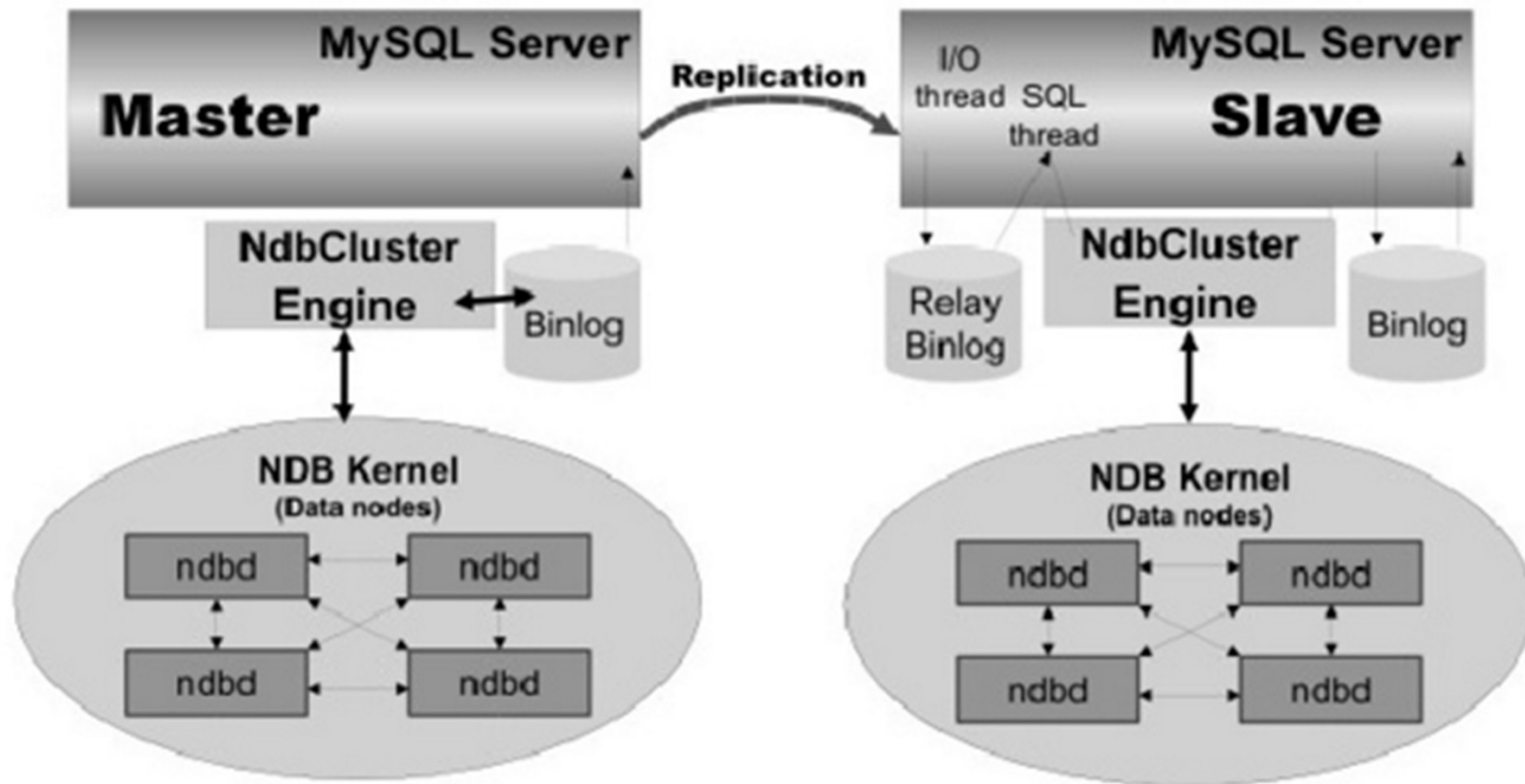


MySQL Cluster





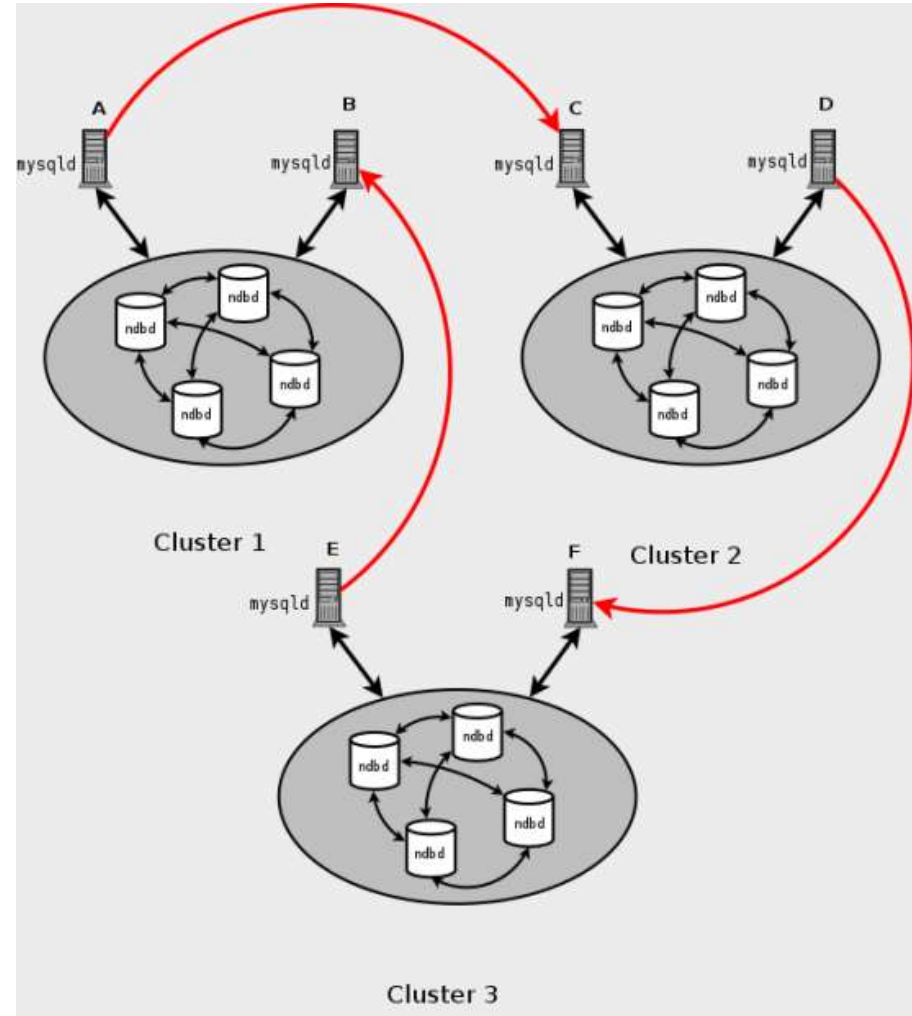
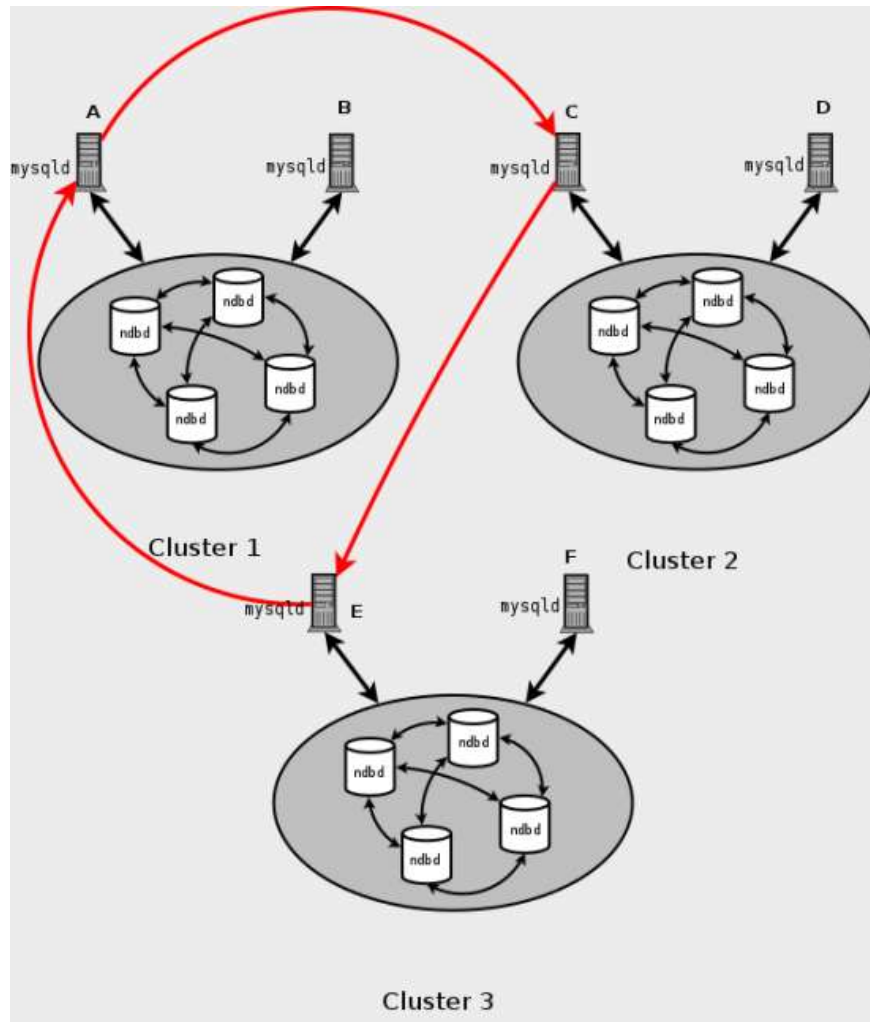
MySQL Cluster



Replication of Clusters



MySQL Cluster



Circular Replication of Clusters



MySQL Cluster

- **Cluster's components;**

- Management Node
- SQL/API Node
- Data/Storage Node

- **Configurations File;**

- Local
- Global
- Connect String

- **Cluster Partitioning**

- Partitions
- Fragments

- **Disk Tables**

- **In-Memory Tables**

- **Replication**

- Synchronous
- Assynchronous
- Replication Channels;

- **Backup & Restore**

- Native Backup
- Native Restore
- mysqldump

- **Performance Tuning**

- Condition Pushdown
- Iptraf